

# Pulsar Signal Classification with Imbalanced Data: Feature Engineering and Multi-Model Comparison on the HTRU2 Dataset

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## Abstract

In this project, our group doing a research on pulsar signal classification, which based on the existing HTRU2 pulsar dataset. In our analysis, the amount of noise data in the dataset was approximately ten times of the amount of real pulsar data, which indicating a huge class imbalance. To solve this problem, we built five models and create four new features, and we use these four features together with the original eight features for model training. The first model, baseline model, we used logistic regression. And then we also trained SVM, Random Forest, Extreme Gradient Boosting, and Multilayer Perceptron model. In these all models, Random Forest model have the best performance. Also, consider the imbalanced data, we used PR-AUC and minority-class recall to tune and evaluate, rather than Conventional accuracy metrics. We compared the importance of all features at the same time, and found that the four new features are ranked highest, which shows that these four features are effectively constructed the optimal boundary.

## 1 Introduction

In this project, our group doing a research on pulsar signal classification, which based on the existing HTRU2 pulsar dataset. In our analysis, the amount of noise data in the dataset was approximately ten times of the amount of real pulsar data, which indicating that the existing pulsar data has a huge class imbalance. Conventional accuracy metrics will be failure when facing this kind of imbalanced data. For example, if we only look at accuracy, and predict all inputs as noise. The accuracy can also exceed 91%, but this model is obviously unusable. Therefore, when we tuning parameters and evaluating models, we uniformly used PR-AUC and minority-class recall as the metrics. To improve the prediction ability and efficiency of the model, our group carried out feature engineering. We used the original 8 features, based on the real physical meaning, create 4 brand new features, and merge these 4 new features with the original 8 features together for model training. We built five models in total. The first model, Logistic Regression as baseline model, gave us some reference. On this basis, we also trained Support

Vector Machine (SVM), Random Forest, Extreme Gradient Boosting (XGBoost) and Multilayer Perceptron (MLP) neural network model. In these all models, Random Forest model have the best performance. At the same time, the 4 new features we created also ranked highest in the feature comparison, which shows that these 4 features effectively help the model build the optimal classification boundary to distinguish signals and noise. In the following parts, we will report in detail our data processing, the parameter tuning process of each model, as well as the final quantitative comparison results and other content.

## 2 Literature Review

### 2.1 Pulsar Signal Classification Background

In existing research, pulsar signal classification usually be seen as a binary classification problem, its main goal is to separate real pulsar signals from noise signals (Lyon et al., 2016). With the development of astronomy, various sky survey projects with different precision and different scope keep producing a large number of astronomical signals. This makes pulsar signal classification, if only rely on manual screening, the efficiency is getting lower and lower, hard to meet the demand. Therefore, more and more researchers and research projects start to use machine learning methods, to complete the automatic classification of whether a signal is pulsar signal (Lyon et al., 2016; Bethapudi & Desai, 2018).

### 2.2 Imbalanced Learning in Astronomical Classification

In the research of pulsar classification problem, there is a very obvious feature on the data, class imbalance. The number of real pulsar samples is far less than the number of noise samples. Therefore, if we only use traditional accuracy as the evaluation metric, it is easy to get misleading results (He & Garcia, 2009). For example, if a method predict most inputs as noise, since the noise accounts for more than 90% of the total, the accuracy of this method will also be very high, even exceed 90%.

But this method is obviously invalid, it cannot really identify the real pulsar signals. At the same time, existing research also shows that, in this kind of imbalanced classification problem, accuracy is not the most suitable metric, while precision, recall, F1-score, as well as precision-recall curve and related metrics may be more referential (He & Garcia, 2009; Saito & Rehmsmeier, 2015). Especially in the pulsar search task, we usually don't want to miss the real pulsar signals, so minority-class recall and PR-AUC can better reflect the classification ability of the model in actual tasks (Lyon et al., 2016; Saito & Rehmsmeier, 2015). This is why in this project, we uniformly use PR-AUC and minority-class recall to do model tuning and evaluation.

### 2.3 Traditional Models for Pulsar Detection

In existing literature, researchers have tried to use various supervised learning models to complete the classification task of identifying pulsars. Early research shows that tree models show high ability in this kind of task, especially when the feature design is reasonable, they can better separate real signals from noise (Lyon et al., 2016). Besides that, there are also studies that compare models like Multi-Layer Perceptron, AdaBoost, Gradient Boosting and XGBoost, and found that different models will show obvious differences under different feature sets and different data distributions (Bethapudi & Desai, 2018). This is why we choose multiple models for training. There are also studies show that ensemble learning methods usually perform better in pulsar classification tasks. This is because models like Random Forest and XGBoost can better handle non-linear relationships, and can learn the complex interactions between different features. Therefore, these models often have strong classification ability in this kind of tabular data tasks (Wang et al., 2019). Based on this research background, in this project we trained Logistic Regression, SVM, Random Forest, XGBoost and MLP, five models in total. We hope to find a more suitable classification method for HTRU2 dataset through comparison. Figures and tables

### 2.4 Feature Engineering and Model Interpretability

Besides the model itself, doing feature engineering is also an important step in pulsar classification research. Existing research points out that

reasonable feature design can effectively improve the model's ability to separate real pulsar signals from noise signals. Original features often cannot fully express the statistical features and physical meaning of the signal itself (Lyon et al., 2016). Therefore, in many pulsar classification studies, feature engineering not only affect the model performance, but also directly affect whether the model can form a clearer classification boundary. On the other hand, feature importance analysis is also often used to explain model results. By comparing the contribution of different features and feature groups to the classification results, researchers can further analyze which features and variables are the most critical in the model, and why the model can achieve better prediction performance (Bethapudi & Desai, 2018). This is quite consistent with our group's research goal. In this project, we based on 8 initial features, combined with certain physical meaning, create 4 new features, then use them together with the original features for model training. This design is not only to improve the prediction ability and efficiency of the model, but also to further verify whether the new features can help the model build a better classification boundary, and the ability of real physical meaning to reflect in the prediction.

### 2.5 Summary of Gaps and Motivation

Overall, according to existing research literature, in the problem of identifying pulsars, the handling of class imbalance, feature engineering, and reasonable evaluation metrics will all have important impact on the model results (He & Garcia, 2009; Saito & Rehmsmeier, 2015; Lyon et al., 2016). At the same time, the performance of different models on pulsar datasets is also different, so it is necessary to compare multiple models (Bethapudi & Desai, 2018; Wang et al., 2019). However, there are still some areas that can be further explored in existing research. First, some studies, although they compare different models, do not always focus on the model's ability to identify the minority class real pulsar signals. Second, some studies only focus on the model results themselves, but do not further analyze whether the newly constructed features can really help the model form a better classification boundary, or take the real physical meaning into account. Based on this research background, this project take HTRU2 dataset as the research object. Under the premise that this dataset itself has severe

class imbalance, combined with feature engineering and multi-model comparison, we uniformly use PR-AUC and minority-class recall as evaluation metrics, further compare feature importance, hoping to more comprehensively analyze the effect of different models and different features in the task of identifying pulsars.

### 3 Dataset and preprocessing

#### 3.1 Dataset Overview

The dataset used in this project is the candidate dataset of HTRU2 pulsar. It contains 17898 observations, each of which represents a candidate signal detected in the pulsar survey. The task is a binary classification problem: label 1 represents a real pulsar, while label 0 represents non-pulsar instances, such as noise or radio frequency interference. In addition, each candidate's target is described by eight original numerical features. The first four features come from the integral pulse profile: Profile\_mean, Profile\_stdev, Profile\_skewness, and Profile\_kurtosis. The other four features come from the DM-SNR curve: DM\_mean, DM\_stdev, DM\_skewness, and DM\_kurtosis.

#### 3.2 Dataset Motivation

Original dataset is suitable for our research questions because it has some characteristics that make the classification task quite challenging. First of all, the dataset is seriously unbalanced. The original data contains 16259 non-pulsar samples, while only 1639 real pulsar samples. This means that the positive class sample accounts for about 9.16% of the data set, while the negative class sample accounts for about 90.84%. This characteristic directly prompts us to put forward a research question on dealing with sample imbalance. In this case, the model can obtain high accuracy by correctly predicting most classes, but it will still miss many real pulsars. Therefore, this task cannot be evaluated only by the accuracy rate. Secondly, features of original dataset could be grouped to profile feature set and DM feature set which produce different aspect on each signal. Then it allows us to apply feature engineering on each feature set. In addition, it prompts us to create new features which include Profile\_cv, DM\_cv, Profile\_shape\_score, DM\_shape\_score which provide the information of relative variance and strength of signal. By comparing mutual information score of all features, we can test which feature set has more contribution, which features

are more importance, whether relative variance and shape information could be useful to detect real pulsars.

Thirdly, the classification boundary is unlikely to be a purely linear boundary. Pulsars are not only recognized by a simple threshold on a single feature. On the contrary, the difference between a pulsar and a non-pulsar depends on the shape and distribution of the signal, especially the skewness and kurtosis of the pulse profile and the DM\_SNR curve. This prompts us to use both linear and non-linear models. Logistic regression provides a simple linear baseline model, while models such as SVM, Random Forest, XGBoost and MLP are used to capture more complex correlations between features. By comparing these models, we can test whether the non-linear decision boundary can provide practical advantages for pulsar detection.

#### 3.3 Data Quality Checking

Before feature engineering, we applied dataset quality check, especially on checking missing instances, duplicate instances, extremely large values, extremely small values, and zero instances. While the result shows that there are no missing values, duplicate instances, or zero instances. And there are no values larger than  $10^6$  and smaller than  $-10^6$ . Those results reflect that the dataset is relatively clear; there is no need to delete or insert instances. Since all features are numeric values, one hot encoding is not required.

#### 3.4 Feature engineering

Features of the original dataset have described some characteristics of the pulse profile and the pulse DM\_SNR curve. However, it does not include the shape and relative variance of the pulse profile and DM\_SNR curve. Relative variance and shape are important since it may distinguish between the real pulser instance and noise/RFI instances. Therefore, we create four features which are Profile\_cv, DM\_cv, Profile\_shape\_score, DM\_shape\_score. include original features and engineered features) could be grouped by Profile feature set and DM feature set.

$$\text{Profile\_cv} = \frac{\text{Profile\_stdev}}{\text{Profile\_mean} + 10^{-6}} \quad (1)$$

$$\text{DM\_cv} = \frac{\text{DM\_stdev}}{\text{DM\_mean} + 10^{-6}} \quad (2)$$

Profile\_cv and DM\_cv describe the coefficient of variances of profile domain and DM domain. They

are calculated by the standard deviation over mean/expected value. Standard deviation could show the the dispersion degree of the emission signal, but it cannot capture dispersion on average level. Two instances might be similar on standard deviation but different on mean value. Then the coefficient of variances becomes crucial since pulser instances and noise/RFI could be different on degree of dispersion at average level.

$$\text{Profile\_shape\_score} = |\text{Profile\_skewness}| + |\text{Profile\_kurtosis}|$$

$$\text{DM\_shape\_score} = |\text{DM\_skewness}| + |\text{DM\_kurtosis}|$$

Profile\_shape\_score and DM\_shape\_score describe the shape of signal in profile domain and DM domain. They are calculated by combination of absolute values of skewness and kurtosis. Skewness tells whether the distribution is symmetry. Kurtosis tells how sharp the peak of distribution. By using the absolute value, it eliminates the effect of direction and focuses on the strength of the signal, since real pulser instance and noise/RFI may generated different strength of signal, then we could use shape information to distinguish between real pulser instance and non-pulser instance

### 3.5 Feature Contribution Analysis

| Feature             | MI Score |
|---------------------|----------|
| Profile_skewness    | 0.225183 |
| Profile_shape_score | 0.205575 |
| Profile_kurtosis    | 0.195685 |
| Profile_mean        | 0.189617 |
| Profile_cv          | 0.169137 |
| DM_shape_score      | 0.121295 |
| DM_stdev            | 0.120651 |
| DM_kurtosis         | 0.114665 |
| DM_mean             | 0.114011 |
| DM_skewness         | 0.113745 |
| DM_cv               | 0.106512 |
| Profile_stdev       | 0.088661 |

Table 1: mutual information score for each featur To determine contribution of detecting pulser instance, we applied mutual information to evaluate the importance of each feature. Other approaches like recursive feature elimination or random forest feature importance rely on model to evaluate the feature importance. Different models may generate different results. Instead, Mutual

information could directly give the dependency between each feature and target label.

The result of table 1 shows the mutual information score for each feature. Profile feature set has the largest contribution since top five features all belongs to profile feature set. Profile\_skewness have the largest MI Score. It indicates that the distributions of real pulser and noise/RFI are different on asymmetry information. Profile\_shape\_score is depending on profile skewness and kurtosis. Second-highest MI Score of Profile\_shape\_score indicates the good contribution shape information. Then combination of absolute value of kurtosis and skewness reflect the uniqueness of pulse shape. In addition, large MI Score of Profile\_cv shows the importance of relative variance. Profile\_stdev has the lowest MI Score. That means standard deviations of real pulser and noise/RFI might be similar. Furthermore, DM feature set has relatively lower MI Score. MI Score of each feature in DM feature set is in the range of 0.1-0.12. Therefore, DM feature set have observable contribution on detecting pulser, but each individual feature is less discriminative that profile features.

In summary, Profile feature set is more important on detecting real pulsers. Profile skewness and kurtosis have largest contributions in original features. Engineered feature Profile\_shape\_socre captures the useful information on pulser detection.

### 3.6 Data Splitting

Before training the models, we applied stratified splitting on engineered dataset that training dataset for 70%, development dataset for 15%, testing dataset for 15%. Training dataset is used for fitting the models, development dataset is used for hyperparameters tuning, testing dataset is applied for model performance evaluation. The splitting ensures fairness on final model evaluation and reduces data leakage. The reason for utilizing stratified splitting is that the dataset is extremely imbalanced. Since the minority class (label 1) might not be evenly distributed in training, development and testing dataset without stratified splitting. It could affect the model performance and final evaluation.

### 3.7 Evaluation

Model evaluation is based on PR-AUC, precision, recall, F1, FP, FN. Accuracy will not be mainly focused on evaluation. Since the dataset is imbalanced, only predicting the minority class (label 0) could achieve high accuracy. PR-AUC is

suitable for binary classification with an imbalanced dataset. It could evaluate whether models can detect real pulsars with the control of false positives. Then it will be the main metric to rank models. Precision will be used to determine how many real pulsars in the positive predictions (model predict class is 1 which is a pulsar). Recall could be used to determine how many real pulsars are successfully predicted. But precision and recall should be considered together with F1, FP, FN. Because models may miss many other real pulsars with high precision, or models may predict pulsar in many noise/RFI cases with high recall. F1 is used to evaluate model performance on balance of precision and recall. FP will report how many noise/RFI instances are predicted as pulsars. FN will report how many pulsar instances are predicted as noise/RFI.

## 4 Methods

### 4.1 Model training setup

After preprocessing and feature construction, we have trained multiple models on the same training set, development set and test set division. The training set is used to fit the model, the development set is used to select hyperparameters, and the test set is only used for final evaluation.

These models include Logistic Regression, SVM using RBF kernels, Random Forest, XGBoost, and a small feedforward multi-layer perceptron (MLP). Logistic Regression, SVM and MLP are trained using StandardScaler, because these models will be affected by the input feature scale. Random Forest and XGBoost use original numerical features for training, because the dependence of tree-based models on feature scales is different from that of Logistic Regression and SVM.

### 4.2 Class imbalance handling

There is a problem of class imbalance in the data set, and only a few real pulsars. Therefore, we have also tested different classes of imbalance handling methods.

For the SVM, we compared three versions: plain SVM, class-balanced SVM and SVM using SMOTE training. SMOTE is only used in the training set. The development set and the test set remain unchanged, so the evaluation results still reflect the original class distribution and do not include the data leakage caused by resampling.

For Random Forest, we use `class_weight="balanced"`. For XGBoost, we use `scale_pos_weight`, which is calculated from the ratio of negative samples to positive samples in the training set. For the multi-layer perceptron (MLP), we tested the regular training settings and the version of random oversampling during the tuning process. The best development PR-AUC value comes from the unsampled MLP version, so this version was finally compared and adopted.

### 4.3 Hyperparameter tuning

The selection of hyperparameters is based on the performance on the development set. The main tuning indicator is average precision, and PR-AUC is used as the summary metric here. This is more appropriate than the accuracy rate, because the positive class sample is scarce, and the model may miss many pulsars even if the accuracy rate is high. The adjustment range is controlled within a reasonable and small range, but it still covers the main parameters of the complexity of the control model. Logistic Regression adjusts the C value, which controls regularization strength. The SVM adjusts the C value and gamma value, which affect the margin and the RBF decision boundary respectively. Random Forest adjusts the number of trees, the maximum depth of trees and the splitting constraints. XGBoost adjusts the number of estimators, tree depth, learning rate and sampling parameters. For MLP, we have adjusted the size of the hidden layer, the regularization intensity, the learning rate and whether sampling has been used.

**Table 1. Final hyperparameter settings selected using the development set.**

| Model               | Main tuned hyperparameters                                   | Best setting                                         |
|---------------------|--------------------------------------------------------------|------------------------------------------------------|
| Logistic Regression | C                                                            | C=0.1                                                |
| SVM                 | C, gamma                                                     | C=10, gamma=0.001                                    |
| Random Forest       | n_estimators, max_depth, min_samples_split, min_samples_leaf | n_estimators=200, max_depth=10, min_samples_split=5, |

|  |                                                                               |                                                                                                                        |
|--|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
|  | n_estimators,<br>max_depth,<br>min_samples_<br>split,<br>min_samples_l<br>eaf | min_samples_le<br>af=2<br>n_estimators=20<br>0,<br>max_depth=10,<br>min_samples_spl<br>it=5,<br>min_samples_le<br>af=2 |
|--|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|

After selecting the best setting, we use the combined data of the training set and the development set to retrain the final model, and then conduct an evaluation on the test set. We recorded the accuracy rate, precision rate, recall rate, F1 score, ROC-AUC, PR-AUC and confusion matrix values. The accuracy rate and ROC-AUC are listed for integrity, but they are not the main comparative indicators. This article mainly discusses PR-AUC, recall rate, F1 score, false positive and false negative. The prediction label is generated using the default decision-making rules of each model, without any additional threshold adjustment.

False negative is particularly important in this task, because false negative means missing the real pulsar candidate target.

## 5 Discussion

In the discussion part, we will do analysis for the following four charts.

figure1-all\_models\_confusion\_matrices

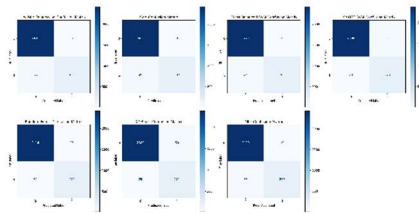


figure2-all\_models\_pr\_curves

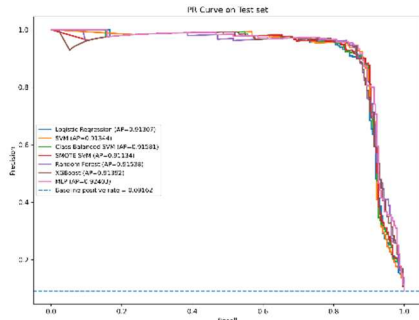


figure3-all\_models\_roc\_curves

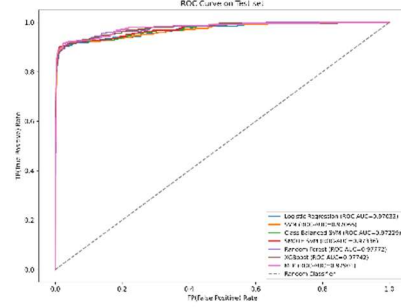
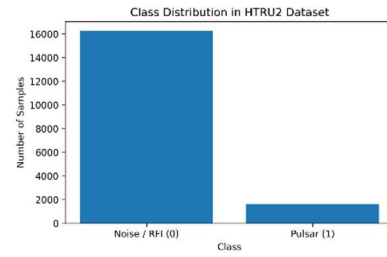


figure4-class\_distribution



### 5.1 Evaluation metric choice under imbalanced data: overcome the misleading of traditional metrics

Combine the class distribution chart figure4, we can see directly. In this task, noise samples are nearly 10 times of pulsar samples. This extreme imbalance will seriously mislead traditional evaluation metrics.

This point got clear proof in curve comparison. In ROC curve figure3, all test models' ROC-AUC are over 0.97, much better than random classifier, look like all work very well. But in PR curve which more fit for imbalanced data figure2, different models' performance difference shows clearly. MLP's PR-AUC reaches 0.924, SMOTE SVM's only 0.911. This gap is almost hidden in ROC.

This also proves that our early choice is reasonable. To overcome traditional metrics' limit in imbalanced data, we give up too much focus on overall accuracy, focus on model's pulsar identification ability, and the balance between precision and recall. This adjustment lets us evaluate model's real value more realistically.

### 5.2 Error pattern analysis: behavior difference of different models and the reasons behind

Through the confusion matrix chart figure1, we break down different models' error patterns, analyze the model and data features behind:

Conservative limit of linear models: Logistic Regression as baseline shows the most extreme error pattern. Its false positive (FP) is only 8, lowest

among all models. But false negative (FN) is as high as 55, missed more than 1/5 of pulsars. Core reason is, Logistic Regression is linear model. The classification boundary between pulsar and noise is not fully linear separable. Many hard samples' features cannot split from noise by linear hyperplane. Also, in imbalanced data, linear model will naturally tend to predict majority class, to reduce overall classification error, make it too conservative to judge positive class. Many edge samples are judged as noise. It is because we realize this limit, so we introduce non-linear models, try to break the limit of linear boundary. Trade-off of imbalance handling methods: For SVM, we test three imbalance handling plans. The result show clear trade-off. Normal SVM's FN is 49. Class balance SVM, by adjusting weight, reduce FN to 46, only increase FP a little. SMOTE SVM, by making samples, reduce FN to 23, least missed detection among all models. But its FP also jump to 59, nearly 6 times of class balance SVM. This shows SMOTE can improve minority class recall effectively, help us reduce missed detection. But its limit is also very obvious. Synthetic samples will blur the decision boundary to some extent, make model have higher tolerance to noise, produce a lot of false detection. In actual use, if you want to not miss any pulsar candidate, SMOTE is okay. If you want to reduce later manual check work, class balance plan is better. Performance advantage of non-linear models: The performance of tree models and neural network, prove that our introduction of non-linear models is effective. Random Forest, as ensemble tree model, by balancing class weight and non-linear boundary, achieve balanced error distribution. FN is 31, FP is 23. It not have too much missed detection, nor too much false detection. This is because ensemble model can capture complex feature interaction, and adapt to imbalanced data. And MLP achieve the best overall performance. Its FP is only 10, FN is 38. It achieve the best balance between precision and recall among all models, can separate those hard samples that linear models cannot handle

### 5.3 Value of feature engineering: fill the information gap of original features

Besides model adjustment, our feature engineering fix the limit of original features. Original 8 features only can describe basic statistics, cannot express the key signal info to separate pulsar and noise. The 4 new features we build have high prediction

contribution, fill the information gap, help model build clearer classification boundary, improve classification ability.

### 5.4 Limit of the research and future direction

This research still have some limits.

Even the best MLP model, still have 38 missed pulsars and 10 false detection. Some samples even with feature engineering and non-linear models cannot separate fully.

All experiments are based on HTRU2 dataset. The methods' generalization need further test on other datasets with different distribution.

Current imbalance handling methods' effect is limited for more extreme cases. In future, we can try more advanced methods, to improve model's performance further.

## 6 Results

Table 2 gives the final test results. The table is kept compact by showing the metrics that are most relevant for this imbalanced binary classification problem.

**Table 2. Compact final test results. Positive class is class 1, real pulsar.**

| Model               | PR-AUC | Precision | Recall | F1     | F | F |
|---------------------|--------|-----------|--------|--------|---|---|
| Model               | PR-AUC | Precision | Recall | F1     | P | N |
| Logistic Regression | 0.9131 | 0.9598    | 0.7764 | 0.8584 | 8 | 5 |
| Logistic Regression | 0.9131 | 0.9598    | 0.7764 | 0.8584 | 8 | 5 |
| SVM                 | 0.9134 | 0.9563    | 0.8008 | 0.8717 | 9 | 4 |
| SVM                 | 0.9134 | 0.9563    | 0.8008 | 0.8717 | 9 | 9 |
| SVM Class Balanced  | 0.9158 | 0.9524    | 0.8130 | 0.8772 | 1 | 4 |
| SVM Class Balanced  | 0.9158 | 0.9524    | 0.8130 | 0.8772 | 1 | 4 |
| SVM SMOTE           | 0.9113 | 0.7908    | 0.9065 | 0.8447 | 5 | 2 |
| SVM SMOTE           | 0.9113 | 0.7908    | 0.9065 | 0.8447 | 9 | 3 |

|                                              |                          |                          |                          |                          |                            |                  |
|----------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|----------------------------|------------------|
| Rando<br>m<br>Forest<br>Rando<br>m<br>Forest | 0.91<br>54<br>0.91<br>54 | 0.903<br>4<br>0.903<br>4 | 0.87<br>40<br>0.87<br>40 | 0.88<br>84<br>0.88<br>84 | 2<br>3<br>2<br>3<br>2<br>3 | 3<br>1<br>3<br>1 |
| XGBoo<br>st<br>XGBoo<br>st                   | 0.91<br>39<br>0.91<br>39 | 0.789<br>3<br>0.789<br>3 | 0.89<br>84<br>0.89<br>84 | 0.84<br>03<br>0.84<br>03 | 5<br>9<br>5<br>9           | 2<br>5<br>2<br>5 |
| MLP<br>MLP                                   | 0.92<br>40<br>0.92<br>40 | 0.954<br>1<br>0.954<br>1 | 0.84<br>55<br>0.84<br>55 | 0.89<br>66<br>0.89<br>66 | 1<br>0<br>1<br>0           | 3<br>8<br>3<br>8 |

Although the accuracy rate is calculated, for this data set, the amount of information about the accuracy rate itself is not large. Since most samples are not pulsars, the model can obtain a high accuracy rate by correctly classifying most classes, but it will still miss a large number of real pulsars. Therefore, the following comparison mainly adopts PR-AUC, recall rate, F1 score and FP/FN trade-off.

The multi-layer perceptron (MLP) model achieves the best overall performance. Its PR-AUC value is the highest, reaching 0.9240, and the F1 score is also the highest, reaching 0.8966. In addition, its number of false positives is also very low, with only 10 false positives on the test set. This makes it the most balanced model in the final comparison. However, its recall rate is only 0.8455, so it is not the best model if the target is just to capture as many pulsars as possible.

The recall rate of SMOTE SVM algorithm is the highest, reaching 0.9065, and the missing rate is the lowest, missing only 23 real pulsars. If the goal is to reduce the number of missed pulsars, it is undoubtedly beneficial. However, the price is also higher: 59 false positives were produced. In practical application, this means that more non-pulsar candidate targets will be sent for further review. This result shows the expected trade-off using the SMOTE algorithm: the recall rate of minority class is higher, but the false positives is also higher.

The class-balanced SVM is slightly improved compared with plain SVM. The recall rate increased from 0.8008 to 0.8130, and the PR-AUC increased from 0.9134 to 0.9158. Although the improvement is smaller than the SMOTE version, it achieves this goal without significantly increasing the false positives.

According to the F1 score, the Random Forest is the strongest among all non-neural network models, with an F1 score of 0.8884, 31 false negatives and 23 false positives. The recall rate of XGBoost is higher than that of Random Forest, but the number of false positives is also higher, resulting in a low F1 score. This means that XGBoost is more inclined to predict positive class, while Random Forest gives more balanced results.

The Logistic Regression model is the most conservative. It has the smallest number of false positives, only 8, but the number of false negatives is also the largest, reaching 55. This means that it can avoid false alarms very well, but compared with other models, it misses too many real pulsars. In general, considering PR-AUC and F1 scores, MLP is the best choice. SMOTE SVM is only more suitable in scenarios where reducing false negatives is more important than reducing the number of false positives.

## 7 Conclusion

Overall, in this project, we doing a research on the pulsar signal problem with HTRU2 dataset. The data has very unbalanced classes. Real pulsar samples are much less than non-pulsar samples (that is, noise). So only accuracy is not enough. It cannot be a good standard to check this task. So we use PR-AUC and minority-class recall as the main standard. We use them to adjust models and compare them. To make the classification better, we use the 8 old features. We look at their real meaning, then build 4 new features. We also train 5 models. They are Logistic Regression, SVM, Random Forest, XGBoost and MLP. In these all models, Random Forest work the best among old machine learning ways. MLP got the highest PR-AUC and F1 score on the test data. SMOTE SVM got the highest recall. But it also made more wrong positive results. This shows we need to balance recall and precision when we work on unbalanced data. Feature analysis also shows that. The 4 new features we made rank high in the comparison. Their real meaning can help the model. It can help build a better line between pulsar signals and noise. Our results shows that. Feature engineering and the choice of check standard are very important for this task. In future work, better ways to handle unbalanced data, and deeper neural network models, can help us do better in finding the minority class.



## Reference List:

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